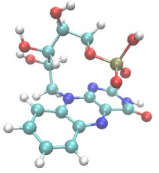
Ground state calculation (external program)



Neglect of XCF + CNDO

Monopole approximation, no RSH = **sTD**

OR

2-center ERI approximation, RSH available = **XsTD**

Approximations

Active MO space

Selection of active MO space

P-CSFs S-CSFs

$N_{CSFs} = N_{P\text{-}CSFs} + N_{S\text{-}CSFs} \ll N_{exci}$

Selection of CSFs

N_{CSFs}

N_{CSFs}

$(A'+B')_{ia,jb}$

Build A'±B' supermatrices

Solve **excitation / response** equation

Final spectrum/property

LR

α(-ω;ω)

ω

OR

ω

Residues of LR

1PA

ECD

QR

β(-2ω;ω,ω)

2ω

Residues of QR

2PA

ESA